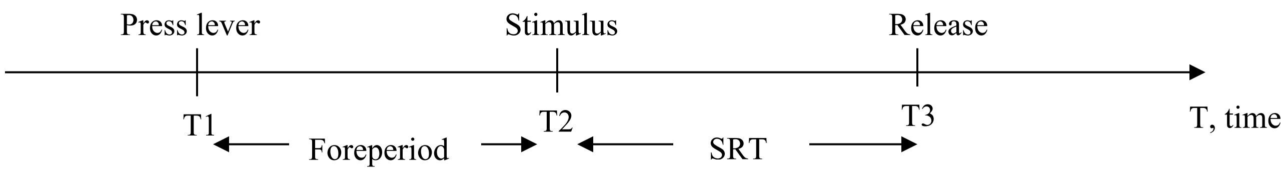
February 04

Training Rats – Protocol

TASK: The rat needs to release the lever as soon as a stimulus (i.e. tone/vibration) is received.

1) Introduction



The main variable in such a task is the response time (RT). The response time is defined as the time between the onset of the stimulus to the release of the lever. This is called a simple response time (SRT) because there is no choice. In other words, all the rat has to do is respond (i.e. tell you there was a stimulus). In a choice-reaction time task, the rat would not only have to tell you there was a stimulus but also discriminate one stimulus from another (i.e. determine if the stimulus has a high or low frequency). The interval from the moment the lever is pressed to the received stimulus is the foreperiod (FP). Foreperiod is a component of the reaction time task, but without stimuli, the task is a delayed response task and that kind of task has a delay period (DP).

In the task, the rat has to maintain a lever press over a variable interval and release the lever in response to auditory and tactile stimulus with a short interval time. When the lever is pressed and released, water is received. While the rat has no problem learning to respond rapidly to the stimuli, it has difficulties inhibiting the tendency to release the response lever prior to the stimuli. When this happens, the rat makes a premature response. The rat may also release the lever after a stimuli is received, resulting in a late response. As a result of a premature and/or a late response, there is a time-out (TO) in which the cage becomes dark for a couple of seconds and no water is given to the rat. During time-out, the rat needs to learn to wait for the light to come back on before it can resume the task of pressing the lever. However, the rat may become frustrated during a time-out and begin to press the lever continuously which will result in time-out lever presses (TOLP). It's likely that the TOLP value will be high in the beginning of the rat's training. Once the rat is trained, the TOLP value should be in the low hundreds, even less than one hundred. If a trained rat has a high TOLP, just give them a little extra water overnight…they could have just had a bad day.

All rats should get around 15ml of water everyday during the week. On Fridays, rats can get their regular water bottles with unlimited water for the weekend. On Sundays, by noontime, these regular water bottles need to be removed so that come Monday, the rats are a bit thirsty making them motivated to train. They usually drink around 7ml to 8ml of water during their training. You should complete that volume to 15ml using the small water bottles in the animal rooms. You should give the rats their water bottle about 1.5 to 2.5 hours after they have completed their training for the day. Try not to give it to them at the same hour everyday; you don't want them to get use to the idea that they will always get water at a specific hour. You'll know a rat is well trained when it's correct responses are greater than it's premature responses by at least 50%, and late responses are around 10 or less.

2) Steps in Animal Training

Animal handling – Handle all the rats prior to training them. They need to get use to you picking them up, petting them, letting them walk on you, etc. You should handle them for at least 3 days.

Water – Rats need to get use to drinking water from the small tubes. Two nights before you begin training them (you can do this even while you are handling them), give the rats water in the small tubes; fill the tubes to 40ml. Do this for at least 2 nights prior to you beginning your training with them.

Turning on/off the Machines – After you've turned on the power supply, the TDT rack, the MedPC rack, the video monitors and their switching boxes, the pumps and the computer monitors (in that order), activate the computer program called Med-PC IV. A window opens which looks like a spreadsheet. On top of the window there are a couple of icons for you to choose from. The icons you need to familiarize yourselves with are open book (start the session), closed book (close session, abandon data, save data), ▲X (change variable values such as foreperiod, reaction time) and ◇ (start signal, send K (give water manually); icon has blue curvy lines on top of square-like structure). Underneath these icons, you have the option (and it is recommended) to write the subject ID, the group and the experiment. Beneath this, you have a drop down menu with all the protocols. You'll need to pick a protocol and also the appropriate box in which the training will take place and then click OK. When you've completed all training at the end of the day, you'll need to close off all the machines in the following order: MedPC rack, TDT rack, power supply, video monitors and switching boxes, pumps, and finally monitors for the PC's.

Preparing Boxes for Training Sessions – Before every training session, you need to change the water in the water tubes and flush out all the old water from the previous day. Make sure to get rid of the air bubbles in the tubing. If, for some reason, the programs won't load up (i.e. error messages), click on the icon called “zBUSmon” and then “Reset” and “OK”. Once all computer programs are functioning, you’ll need to run the SRT TEST protocol (using MatLab software) in order to calibrate the background noise in each cage (the background noise should be approximately 68dB) and do a quick maintenance check making sure that the water pumps work, the levers aren’t stuck, the drinking tubes aren’t plugged and that there is a stimulus (tone/vibration) after you press each lever a couple of times. Once you’ve completed the SRT TEST, you can begin training the animals. Make sure to write the rat ID and water volume on each cage door so that you’ll know which rat is in what cage and the volume of water they consumed during the session.

PLEASE NOTE: At the end of the day, when all the rats have been trained, you’ll need to clean out the trays from the urine and feces. You can do this by washing the trays in warm water and wiping them dry with paper towels. Also, once a week you’ll need to clean out the cages using a cleaning agent called RoCal. You need to put a little bit of RoCal in the bottle and fill it up with water. Use paper towels (with RoCal solution) to wipe down the cage door, the inner walls and the stage (top and bottom). When you’re done cleaning out the cages, leave the cage doors open throughout the night so that the RoCal smell can evaporate. (Rats don’t like this smell and it may affect their training.)

3) Training Protocols

PLEASE NOTE: All protocols run on MatLab software! Also, make sure to write down all data in a spreadsheet and to save (flush) all data at the end of each training day.

Autoshape – This protocol usually takes one session and it is 30 minutes long. Before you begin this protocol, make sure to remove the lever from the cage! In this training protocol, the rat acclimates to its new environment, new smells, new sounds, and learns that water comes out of the tube. Water is automatically pumped to the waterspout. You should watch the rat on the monitor to make sure that it's all right and that it is drinking from the water tube. The rat should start drinking from the waterspout after a few minutes. If the rat is not behaving, then you can run the rat again in autoshape the next day. If the rat is behaving, it can move on to the PRESS protocol.

Press – In this protocol, the rat needs to learn to press the lever and that when it presses the lever, it is rewarded with water from the water tube. Each session in this protocol should be 60 minutes. Before you begin this protocol, make sure to put the lever right under the water tube. Also, put a few drops of water on the lever. Doing this will temp the rat to the lever. The rat should run on this protocol until its correct responses are more than 50. During this session, you can help motivate the rat by giving the rat water when it is near the lever (in the beginning of the session) and when its paw is on the lever (throughout the session). Once the rat has at least 50 correct responses if not more (this may take a couple of days), you can move it to the RELEASE protocol.

Release – In this protocol, the rat needs to learn to press the lever and release it, in order to get water from the water tube. Each session in this protocol should be 60 minutes. Before you begin this protocol, make sure to put the lever right under the water tube. Also, put a few drops of water on the lever to help motivate the rat. If you notice the rat isn't pressing the lever, you can open the cage, during the session, and put some more drops of water on the lever. But don't do this too often; once or twice is enough. The rat should run on this protocol only once! Even if the rat doesn't behave (unlikely), it needs to be moved to the next protocol which is the WAIT protocol.

Wait Protocol – This protocol has three different parts to it. Wait 1 is the initial protocol for the delayed response in the reaction time task. This should be run until the rat waits for the desired foreperiod. Then run Wait 2 until the rat reaches the desired foreperoid. Finally, run Wait 3 until the rat is doing 50% more correct trials than premature trials. This can take at least one to two weeks. Each session should only be 45 minutes long. Once the rat is running well on Wait 3, it can be moved to the SRT protocol.

SRT Protocol – In this task the rats have to hold down the lever for at least 100ms or else the tone and/or vibration won't be presented. Also, they must release the lever within 2 seconds of the onset of the stimuli. If they do this, they are rewarded with water. After 3 consecutive trials, the "waiting time" or "foreperiod" is increased by 50ms. If they release the lever before the stimuli or wait longer than 2 seconds (the upper limit for the reaction time), they are "punished" with a "time-out" period, where the light turns off and they have to withhold responding for 5-10 seconds. (If they do press during the time-out, the interval is reset.) Each session should only be 45 minutes long.

This protocol has many steps. Training is as follows:

<u>T-SRT1:</u> Rats need to run on this task for 2 sessions. The rule is 3-in-a-row correct to increment the FP from 100ms to the desired foreperiod by 0.05s. Move the rat on to the T-SRT2 protocol regardless if it does or doesn't reach the desired foreperiod by the end of the second session.

<u>T-SRT2:</u> Rats need to run on this task for 2 sessions. Rule is 2-in-a-row correct and FP increases by 0.1sec from 100ms to the desired foreperiod.

T-SRT3: Rats need to run on this task for 2 sessions. Rule is 1-in-a-row correct to go from 100ms to the desired foreperiod in 100ms; FP is jittered by +/-0.25ms once the rat gets to 1sec.

T-SRT4: Rats need to run on this task for 7 sessions. Rule is 1-in-a-row correct to go from 100ms to the desired foreperiod in 100ms; FP is jittered by +/-0.25ms once the rat gets to 1sec.

T-SRT5: Rats need to run on this task for 15 sessions. Rule is 1-in-a-row correct to go from 100ms to the desired foreperiod in 100ms; FP is jittered by +/-0.25ms once the rat gets to 1sec.

T-int: Rats need to run on this task for 3 consecutive days. At the end of the third day, the rat has completed all training.